

Table of contents

Conjecture of ?@eq-moneyquant and ?@eq-workthermal

How consistently map economic measures to the quantities of nature to model the economical system?

Input:

$$Input = f_0(Resouces, Labor, Capital) = \text{Work} = \Delta \text{Energy} = p \cdot v \quad (1)$$

Output:

$$Output = \text{Goods} + \text{Services} = \text{Production} = \Delta \text{State} = V_I \quad (2)$$

Productivity:

$$Productivity = \frac{Output}{Input} = \frac{\Delta State}{\Delta Energy} = \frac{Production}{Work} = \frac{V_I}{pv} \quad (3)$$

see description of “movement” ?@eq-movement

Table 1: Core Measurements

Economy	→	←	Physics
Resources	Assets	Force	Matter
Capital	Liabilities	Momentum	Space
Labor	Expenses	Work	
	Income	Energy	Time
		Production	

As a first approach, considering the geometrical perspective of the concept of “work” (?@eq-workgeom), applied to economical measurements, so to satisfy the following system:

$f_A(Assets) = f_B(Liabilities + Gain) = \text{“momentum” } p$ from initial state r_0 to new state r_1 , where the change of state is $r = r_1 - r_0$

Matter substance, object movement/change:

$$A = f_A = F \cdot t = p \quad (4)$$

Space of movement, maneuverability:

$$B + R = f_B = m \cdot v = p \quad (5)$$

Work input (efforts, costs) through labor force, machines and infrastructure applied, and materials consumed for production:

$$W = f_0 = F \cdot r = m \cdot \frac{v}{t} \cdot r = p \cdot v \quad (6)$$

Energy transfer, energy change, profit or loss:

$$Y = \Delta E = \frac{1}{2}pv + \mathcal{U} \quad (7)$$

Production value inflation adjusted:

$$P \cdot V_I = \sum Y_{adjust} = \sum (W + R)_{adjust} = \Delta U - \Delta H = Q \cdot N \quad (8)$$

see [?@eq-workthemoostat](#) and [?@eq-workgeom](#)

Thus, to be determined are v (economy velocity) and p (financial structure). Eventually, determine r = “state change through trajectory” (substance change to the new assets structure) by given v = “velocity of movement” (Business speed = monetary value, price with inflation), or determine v by given r . The individual positions $\{i, j, l, m\}$ of Liabilities $\{B_i + R_j\}_k$, Assets $\{A_m\}_k$, and Expenses $\{W_l\}_k$ shall be grouped in sets $\{k\}$, so to satisfy $f_k(i, j) = f_k(m) = f_k(l) = \{pv\}_k$.

The constraints shall be modelled with the Lagrangian. Hence, “Work” shall be derived from the Hamiltonian which leads to the identification of the “movement” (= change, transformation) of the trajectory (= path to new state) with $\dot{r} = v$ and $\dot{p} = F$.

Experiment design: **p and v are modelled ex-ante with means of CONSTRAINTS from contractual agreement and from given juristic legislation, in such a way that physics laws are satisfied. Decisions making occure within this ex-ante frame.**